Extension concept from original paper:

Title: "Near-optimal Sensor Placement for Detecting Stochastic Target Trajectories in Barrier Coverage Systems".

Problem Discussion

Kim et al. [1] tackle the problem of deploying a limited number of underwater sensors on the seafloor to detect ships crossing a bounded ocean region. The challenge is that future ship trajectories are unknown, and only historical AIS tracking data is available.

The core idea is to reformulate the geometry of the problem. Any straight-line ship trajectory crossing the region can be uniquely represented using two parameters, (i) its angle α and (ii) its perpendicular distance p from the origin. This maps each trajectory from physical "inertial space" into a point in a parameterized "representation space" (α, p) [2]. Using historical AIS data, a density function $\lambda(\alpha, p)$ is then learned over this space using a log-Gaussian Cox process (LGCP), fitted via the INLA method [3,4]. This provides a probabilistic estimate of traffic intensity while also accounting for uncertainty in the observations.

Sensor detection is modeled as a distance-dependent process which each sensor has a detection probability that decreases with the minimum distance between the sensor and a ship's trajectory. With multiple sensors, a ship is only missed if all sensors fail to detect it, so the overall miss probability is the product of the individual miss probabilities.

The optimization objective is formulated as the void probability, which the probability that no ships pass through the region undetected. This is computed by integrating the product of the intensity $\lambda(\alpha, p)$ and the joint miss probability over the representation space, followed by applying a negative exponential. Intuitively, well-placed sensors reduce missed detections in high-density regions, effectively "thinning" the intensity and increasing the void probability toward 1.

Due to the optimal placement problem is NP-hard [5], the authors rely on the submodularity of the objective, which guarantees diminishing returns as more sensors are added. This structure enables a greedy placement strategy that sequentially selects sensor locations with the largest marginal gain, achieving at least $(1 - 1/e) \approx 63\%$ of the optimal solution [6]. The resulting configuration is then further improved using Newton-type refinement methods [7].

A key limitation of this framework is that it assumes a single stationary traffic intensity function. In reality, maritime traffic is dynamic and varies with season, weather conditions, and between day/night. For instance, A deployment tuned to January traffic may fail in July. As a result, a deployment optimized for one traffic pattern may have blind spots during another. This motivated my extension.

Proposed Extension Study:

I propose robust sensor placement under non-stationary target intensity. Instead of assuming one fixed intensity $\lambda(l)$, I consider K distinct intensity scenarios $\lambda_1(l)$, $\lambda_2(l)$, ..., λ_K(l), where each scenario represents a different maritime traffic regime, as illustrated in Figure 1. For instance, calm-season channel traffic versus storm-season dispersed routing versus rerouted traffic during a port construction event. The core problem is to determine a single fixed sensor deployment that performs reliably across all (K) traffic scenarios simultaneously. Unlike mobile sensing systems, underwater sensors deployed on the seafloor cannot be repositioned easily once installed. Therefore, the sensor configuration must remain effective even when traffic patterns shift over time.

To the best of my knowledge, existing barrier coverage and underwater sensing studies do not address robust optimization under multiple Cox-process intensity scenarios. Prior work generally assumes a single known target intensity function [1] or focuses on other uncertainties, such as sensor failures or communication constraints, rather than uncertainty in the traffic distribution itself.

My formulation connects underwater barrier coverage with the broader fields of robust optimization. This creates a new research direction that is both theoretically interesting and practically relevant for long-term maritime monitoring in dynamic environments.

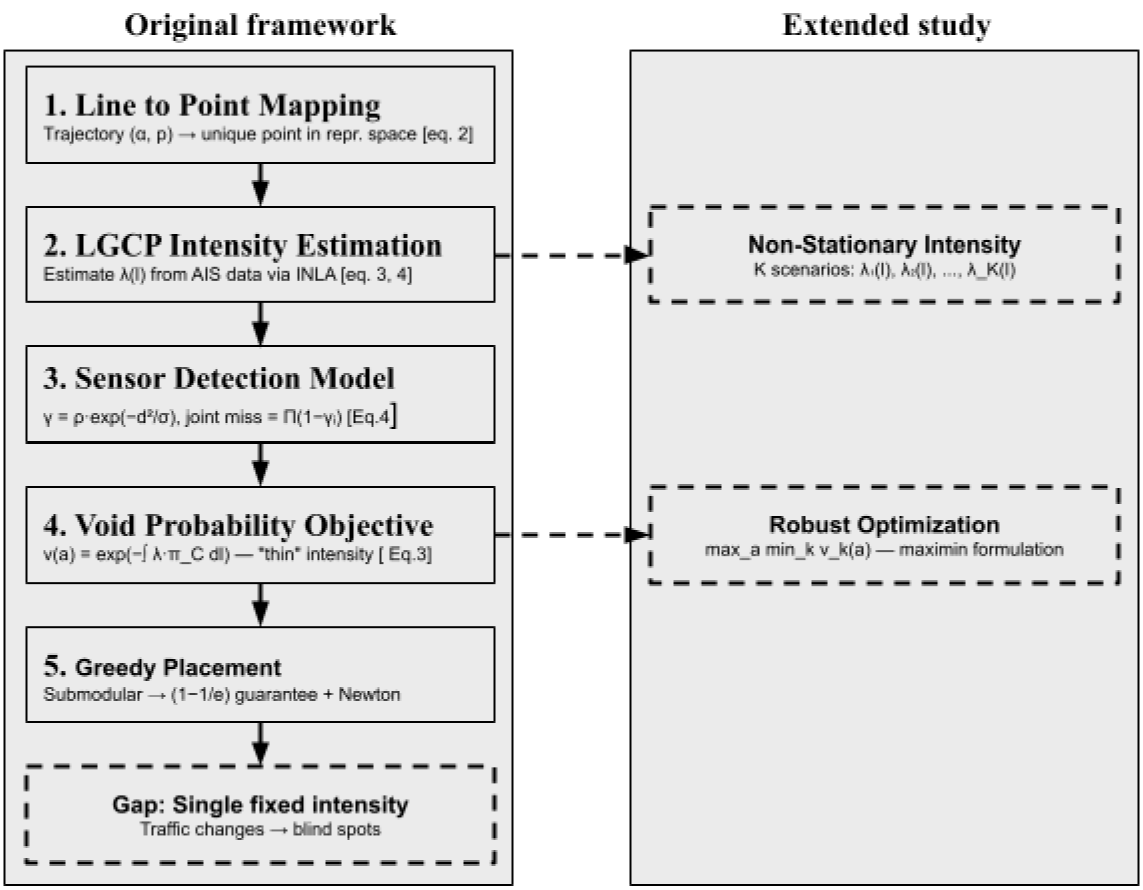

Figure 1: Flow chart of extended study and original framework

For a more detailed understanding of the proposed framework, I would like to implement and simulate the method step by step in **Jupyter Notebook (source code)**. This will allow the mathematical formaulation, experiment, and sensor placement behavior under different traffic scenarios to be visualized and analyzed interactively.

References

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